HEALTH MONITORING NODE PROGRAM-2

Name : Poruri Nageswara Sumanth

Reg No:192411082

➔ **Source Code:**

```
#include <Wire.h>

#define LM75_ADDRESS 0x48

#define PULSE_PIN 8

unsigned long pulseCount = 0;

unsigned long lastTempTime = 0;

unsigned long lastBpmTime = 0;

int previousPulseState = LOW;

float readTemperature()

{

  Wire.beginTransmission(LM75_ADDRESS);

  Wire.write(0x00);

  Wire.endTransmission();
```

```
  Wire.requestFrom(LM75_ADDRESS, 2);

  if (Wire.available() >= 2)
  {
    byte msb = Wire.read();
    byte lsb = Wire.read();

    int16_t rawTemperature =
      ((int16_t)msb << 8) | lsb;

    return rawTemperature / 256.0;
  }

  return -999.0;
}

void setup()
{
  Serial.begin(9600);
  Wire.begin();

  pinMode(PULSE_PIN, INPUT);
```

```
  Serial.println("I2C HEALTH MONITORING NODE");
  Serial.println("---------------------------");
}

void loop()
{
  unsigned long currentTime = millis();

  /* Pulse counting */

  int currentPulseState = digitalRead(PULSE_PIN);

  if (currentPulseState == HIGH &&
      previousPulseState == LOW)
  {
    pulseCount++;
  }

  previousPulseState = currentPulseState;

  /* Temperature acquisition every 1 second */

  if (currentTime - lastTempTime >= 1000)
```

```
{
  lastTempTime = currentTime;

  float temperature = readTemperature();

  Serial.print("Temperature: ");

  if (temperature > -100)
  {
    Serial.print(temperature, 1);
    Serial.println(" C");
  }
  else
  {
    Serial.println("Sensor Error");
  }
}

/* BPM calculation every 5 seconds */

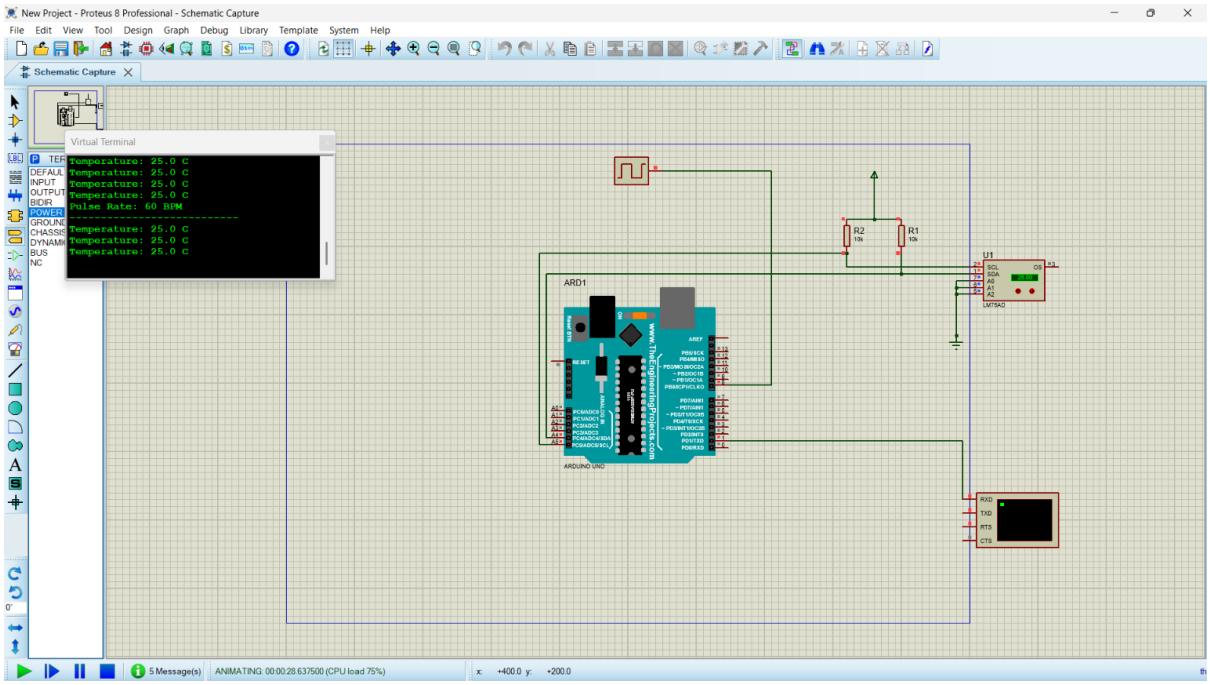
if (currentTime - lastBpmTime >= 5000)
{
  lastBpmTime = currentTime;
```

```
    float bpm = (pulseCount * 60.0) / 5.0;


    Serial.print("Pulse Rate: ");
    Serial.print(bpm, 0);
    Serial.println(" BPM");


    Serial.println("----------------------------");


    pulseCount = 0;
  }
}
```



Reflections:

- The LM75 temperature sensor measures temperature and communicates with the microcontroller using the I²C protocol.
- Temperature readings are obtained every 1 second and displayed on the Serial Monitor.
- The pulse sensor detects rising-edge signals, and the pulse rate is calculated in BPM every 5 seconds.
- The millis() function enables periodic temperature measurement and pulse monitoring without blocking delays.
- The program displays sensor errors when temperature data is unavailable and demonstrates the basic working of a health monitoring node.